Review plan · for the teacher

Python for teachers: unit check

Programming · Level 2 · Unit check

Time	Materials
45 min	Board, chalk, paper. No devices needed.

Learning goals

- First programs
- Decisions
- Loops
- Lists

Activity: Team quiz

1. Make teams of four
2. Read each question aloud; teams agree on one answer
3. Reveal the answer and ask one team to explain why
4. Write the hardest question on the board for tomorrow

Discussion

- Which program will you write first for your own work?
- Which question surprised you, and why?
- Which idea will your pupils find hardest?

Exit ticket

- One job I will hand to Python this month, and how

Try it at school tomorrow
Pick one job from your week (averages, a register count, grade bands) and write the program for it on your phone. Then show it to a colleague.

Quiz sheet · for each team

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Team _____ Date _____

1 **What does this program print?**

```
x = "4"
y = 2
print(x * y)
```

◯ 44 ◯ 8 ◯ An error

2 **When a pupil types 15, input() gives back the number 15.**

◯ True ◯ False

3 **Put the lines in order: the program finds the top mark in the list without using max(). One line doesn't belong.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `best = marks[0]`

☐ `if m > best:`

☐ `if m < best:`

☐ `marks = [14, 9, 17, 11]`

☐ `print("Top mark:", best)`

☐ `best = m`

☐ `for m in marks:`

4 **Match each line to what it prints.**

`print(7 * 6)`	• • • •	3
`print("7 * 6")`	• • • •	2.5
`print(len([4, 8, 1]))`	• • • •	42
`print(10 / 4)`	• • • •	7 * 6

5 **Fill the gaps: the program saves 15 a week and counts the weeks until the total reaches 100.**

```
total = 0
weeks = 0
while total _____ 100:
    total = total + 15
    weeks = weeks _____ 1
print(weeks)
```

Word bank: `>` `-` `<` `+`

6 **Python stops this program with a SyntaxError. Which line is wrong?**

Circle the line with the mistake. Write it correctly:

```
1  mark = int(input("Mark? "))
2  if mark >= 50
3      print("Pass")
```

7 **What does this program print?**

```
mark = 85
if mark >= 50:
    print("B")
elif mark >= 70:
    print("A")
```

◯ A and B ◯ B ◯ A

8 **Which are true for marks = [14, 9, 17]?**

☐ marks[3] is 17 ☐ len(marks) is 3 ☐ marks[0] is 14 ☐ marks[-1] is 17

9 **Put the debugging routine in order.**

☐ Go to the line the message names

☐ Run the program and read the error message

☐ Run it again to check

☐ Change one thing

10 **Finish the program: it asks for a number and prints the total of the whole numbers from 1 up to it. For 10 it prints 55.**

```
n = int(input("Up to? "))
total = 0
# add up 1 to n, then print the total
```

Write your program:

11 **A while loop always runs at least once.**

◯ True ◯ False

Answers · for the teacher

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1 **44**

x is the text "4", and text times 2 repeats it: "44".

2 **False**

input() always gives back text, here "15".

3 **Top to bottom: 2, 4, X, 1, 6, 5, 3**

```
marks = [14, 9, 17, 11]
best = marks[0]
for m in marks:
    if m > best:
        best = m
print("Top mark:", best)
```

Start with the first mark as the best so far, then let every bigger mark replace it.

4 **print(7 * 6) — 42**
print("7 * 6") — 7 * 6
print(len([4, 8, 1])) — 3
print(10 / 4) — 2.5

Sums are worked out; text in quotes is shown as written; len counts the items; / gives a decimal.

5

```
total = 0
weeks = 0
while total < 100:
    total = total + 15
    weeks = weeks + 1
print(weeks)
```

The loop runs while the total is still below 100, and each turn adds one week.

6 **Line 2: `if mark >= 50:`**

A line that starts a block (if, elif, else, for, while, def) must end with a colon.

7 **B**

85 is 50 or more, so the first branch runs and prints B; the elif is never checked.

8 **marks[0] is 14; len(marks) is 3; marks[-1] is 17**

Lists count from 0, so the three marks are at 0, 1 and 2, and -1 means the last.

9 **Top to bottom: 2, 1, 4, 3**

Read the message first: it names the kind of error and the line.

10

```
n = int(input("Up to? "))
total = 0
for i in range(1, n + 1):
    total = total + i
print(total)
```

A running total: start at 0 and add each number.

11 **False**

while checks its condition before every turn, the first one included.